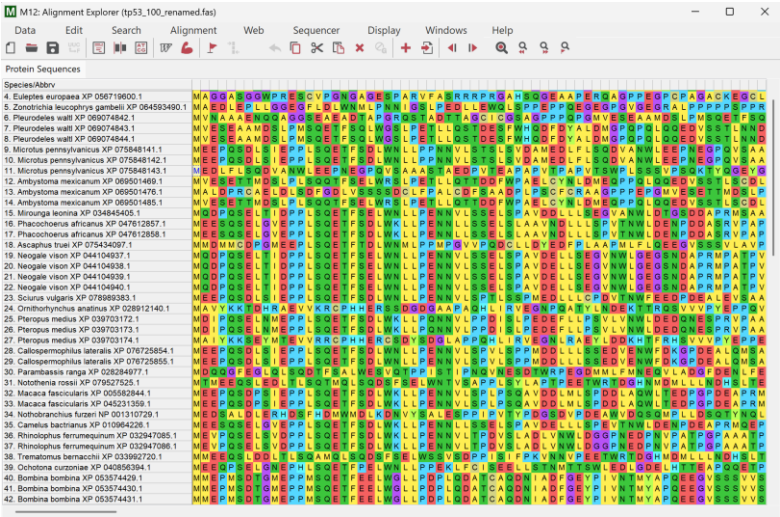
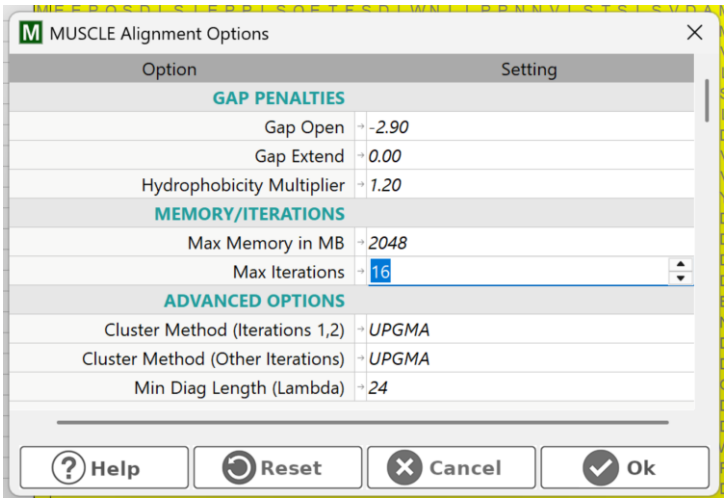
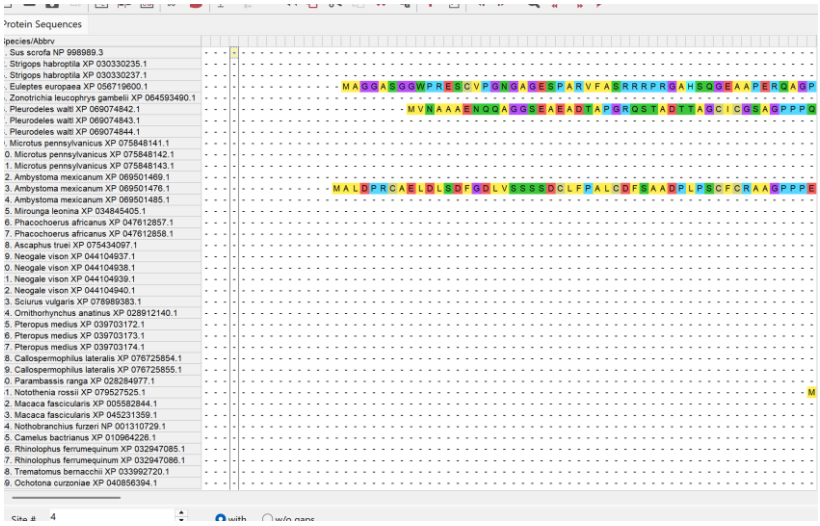
Задание 2

Откроем наш файл в программе MEGA12 (пришлось предварительно изменить формат файла на .fas, чтобы программа открыла его).



Сделаем множественное выравнивание (MSA) подготовленных последовательностей.





Также выровняла с T-Coffee

T-Coffee alignment result

MSA

The multiple sequence alignment result as produced by T-coffee.

```
T-COFFEE, Version_11.00 (Version_11.00)
Cedric Notredame
SCORE=884
*
 BAD AVG GOOD
*
Sus scrofa NP 9    :  89
Strigops habrop    :  90
Strigops habrop    :  91
Euleptes europa    :  87
Zonotrichia leu    :  89
Pleurodeles wal    :  87
Pleurodeles wal    :  87
Pleurodeles wal    :  87
Microtus pennsy    :  89
Microtus pennsy    :  89
Microtus pennsy    :  91
Ambystoma mexic    :  87
Ambystoma mexic    :  87
Ambystoma mexic    :  87
Mirounga leonin    :  89
Phacochoerus af    :  89
Phacochoerus af    :  89
Ascaphus truei     :  86
Neogale vison X    :  89
Neogale vison X    :  89
Neogale vison X    :  89
Neogale vison X    :  89
Sciurus vulgari    :  89
Ornithorhynchus    :  93
Pteropus medius    :  89
Pteropus medius    :  89
Pteropus medius    :  93
Callospermophil    :  89
Callospermophil    :  89
Parambassis ran    :  87
Notothenia ross    :  87
Macaca fascicul    :  89
Macaca fascicul    :  89
Nothobranchius     :  88
Camelus bactria    :  89
Rhinolophus fer    :  89
```

```
Sus_scrofa_NP_9    MEE-------------------------------------------------------
Strigops_habrop    M--------------------------------------------------------
Strigops_habrop    M--------------------------------------------------------
Euleptes_europa    MAG------------------------------------------------------
Zonotrichia_leu    MAE------------------------------------------------------
Pleurodeles_wal    MVNAAAENQQAGGSEA----------EAD-----------TAP-----------GRQS-----
Pleurodeles_wal    MVES-----------------------------------------------------
Pleurodeles_wal    MVES-----------------------------------------------------
Microtus_pennsy    MEE------------------------------------------------------
Microtus_pennsy    MEE------------------------------------------------------
Microtus_pennsy    MED------------------------------------------------------
Ambystoma_mexic    MV-------------------------------------------------------
Ambystoma_mexic    MALDPRCAE-LDLSDFGDLVSSSSDCLFP-----------ALC-----------DFSA-----
Ambystoma_mexic    MV-------------------------------------------------------
Mirounga_leonin    MQD------------------------------------------------------
Phacochoerus_af    MEE------------------------------------------------------
Phacochoerus_af    MEE------------------------------------------------------
Ascaphus_truei     MMD------------------------------------------------------
Neogale_vison_X    MQD------------------------------------------------------
Neogale_vison_X    MQD------------------------------------------------------
Neogale_vison_X    MQD------------------------------------------------------
Neogale_vison_X    MQD------------------------------------------------------
Sciurus_vulgari    MEE------------------------------------------------------
Ornithorhynchus    -------------------------------------------------------
Pteropus_medius    MDI------------------------------------------------------
Pteropus_medius    MDI------------------------------------------------------
Pteropus_medius    -------------------------------------------------------
Callospermophil    MEE------------------------------------------------------
Callospermophil    MEE------------------------------------------------------
Parambassis_ran    MDQ------------------------------------------------------
Notothenia_ross    MTM------------------------------------------------------
Macaca_fascicul    MEE------------------------------------------------------
Macaca_fascicul    MEE------------------------------------------------------
Nothobranchius_    MED------------------------------------------------------
Camelus_bactria    MEE------------------------------------------------------
Rhinolophus_fer    MEV------------------------------------------------------
Rhinolophus_fer    MEV------------------------------------------------------
Trematomus_bern    MME------------------------------------------------------
Ochotona_curzon    MEE------------------------------------------------------
Bombina_bombina    MME------------------------------------------------------
Bombina_bombina    MME------------------------------------------------------
Bombina_bombina    MME------------------------------------------------------
Myotis_daubento    MDE------------------------------------------------------
Myotis_daubento    MDE------------------------------------------------------
Myotis_daubento    MDD------------------------------------------------------
Myotis_daubento    MDD------------------------------------------------------
```